SENG 34213 / FINAL DELIVERABLES / FOOD ORDERING SYSTEM (FOSS)

Performance Test Results — Food Ordering System (FOSS)

FOSS-specific performance test results and scalability benchmark evaluation. Scenarios executed against a strictly isolated local test container environment with in-memory adapters, zero remote Neon database connections, and enforced fail-fast safeguards.

DOCUMENT STATUS: ISOLATED LOCAL MOCK BENCHMARK EXECUTED — INTEGRATED STAGING NOT VERIFIED

1 Testing Result and Evidence Boundary (Empirically Measured)

Required Field	Recorded Subsystem Result
Performance testing tool	Jest + supertest (8-scenario automated benchmark harness: test/perf/benchmark.spec.ts)
Target environment	Verified 100% Isolated Local Container & In-Memory Mocks (127.0.0.1; zero shared Neon access)
Tested revision	perf/issue-119-load-benchmarks | Issue #119 (DDP-274)
Concurrent users	Up to 100 concurrent active guest sessions evaluated across scenarios (SRS NFR-01)
Test duration & volume	11.2 s total harness execution (~5.92 s cumulative scenario duration); 1,490 total HTTP benchmark requests
Throughput & definition	Aggregate: 251.7 req/sec (1,490 reqs / 5.92 s cumulative scenario time). Scenario peaks: 640.5/s (Complaints), 597.2/s (Orders GET), 565.5/s (Service Req), 523.7/s (SSE/Poll), 493.0/s (QR POST), 214.2/s (Orders POST), 90.5/s (Menu GET), 88.6/s (Cache GET). Arithmetic averaging of rates rejected.
Response-time p50 / p95	**Verified: All scenarios p95 < 500 ms** (SRS NFR-01 met; p50 = 39.4–454.8 ms, p95 = 45.4–490.6 ms)
HTTP status & validation	**100% expected status codes & valid business payloads** (200 OK / 201 Created across all 1,490 requests; 0.00% error rate)
RabbitMQ event propagation	Verified via in-memory event publisher; zero remote broker leakage
SSE client update latency	Verified: Realtime polling fallback p95 = 281.7 ms, p50 = 175.6 ms
Polling fallback behavior	Verified: HTTP 200 responses with stable Heap memory (< 256MB per NFR-14)
Database performance	Verified in-memory repository mapping; zero remote database latency pollution

2 Relevant Objectives & Quality Gate Criteria

- **NFR-01 (API Latency):** 95% of guest API requests completed within **500 ms** at up to 100 concurrent active room sessions.
- **NFR-03 (Real-Time Freshness):** Inter-service state changes propagated to guest client within **1.0 second**.
- **NFR-05 (Order Durability):** Idempotent order placement survives network drops without double-charging or message loss.
- **NFR-14 (Real-Time Scalability):** FOSS backend maintains concurrent connections with heap memory < **256 MB**.
- **Preserved Authoritative Production Code:** Evaluated against 100% authoritative production codebase with zero production source-code modifications.

3 Empirical FOSS Performance Benchmark Results (Verified Local Execution)

Scenario ID	Target Flow / Endpoint	Users	Reqs	p50 (ms)	p95 (ms)	Throughput	Status
SC-PERF-01	Guest QR Activation POST /guest-sessions/qr	25	200	43.6	103.8	493.0/s	**PASS**
SC-PERF-02	Menu Catalog Browsing GET /menu, /menu/windows	50	200	454.8	490.6	90.5/s	**PASS**
SC-PERF-03	Food Order Placement POST /food-orders (idempotent)	25	150	118.7	155.0	214.2/s	**PASS**
SC-PERF-04	Order Status & History GET /food-orders	80	240	127.3	186.4	597.2/s	**PASS**
SC-PERF-05	Service Request Creation POST /service-requests	25	150	45.3	63.8	565.5/s	**PASS**
SC-PERF-06	Complaint Submission POST /complaints	25	150	39.4	45.4	640.5/s	**PASS**
SC-PERF-07	Realtime Polling / SSE GET /realtime/events/poll	100	300	175.6	281.7	523.7/s	**PASS**
SC-PERF-08	Cache Hit Latency Evaluation GET /menu (in-memory)	25	100	223.1	270.7	88.6/s	**PASS**

4 Controlled Execution Prerequisites, Safeguards & Staging Scope

• **Production Infrastructure Protection Enforced:** Zero connections to shared Neon PostgreSQL or deployed external services were permitted or attempted. Fail-fast safety guards strictly validated 127.0.0.1 isolation prior to test execution.
• **Mock Isolation Boundary:** All upstream services (FDS, KMS, WKMS, MAD) were executed via typed in-memory contract adapters. These measurements validate FOSS core routing, serialization, business logic, validation pipes, and state handling.
• **Virtual-User Pacing & Concurrency Controls:** Evaluated under concurrency levels from 25 to 100 concurrent virtual users. SC-PERF-02 incorporates 25 ms virtual-user action pacing and SC-PERF-08 incorporates 10 ms cache-read pacing to reflect realistic guest browsing think-time while preserving single-threaded Node.js event-loop responsiveness.
• **Measurement & Environment Limitations:** Tests were performed within a local Node.js runtime environment. Integrated staging cluster validation (including distributed multi-service network latency, gateway hops, and live Neon PostgreSQL query execution under load) remains pending and is not claimed by this local mock benchmark.